DO NOW — RECALL

Denary → floating point

Name: _____ Class: _____ Date: _____

1. Choose a type

2. Convert a hexadecimal byte

3. Use two’s complement

4. Explain a fixed-width shift

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Real / float: a number with a fractional part.

2. Model answer

00111100 = 32 + 16 + 8 + 4 = 60.

3. Model answer

+26 = 00011010; invert → 11100101; add 1 → 11100110 .

4. Model answer

01100000 = 96. The leading 1 is discarded; 176 × 2 = 352 exceeds the byte’s maximum 255.

By the end, you can…

• convert denary fractions to binary;

• represent positive and negative denary values in normalised floating-point form using an eight-bit two’s-complement mantissa and a four-bit two’s-complement exponent;

• normalise a representation without changing its value.

Vocabulary — know these terms

Recall the term — Mantissa

Recall the term — Exponent

Recall the term — Binary point

Recall the term — Normalisation

Recall the term — Precision

Recall the term — Range

NEW LEARNING

Read, annotate and explain

Fraction · mantissa · exponent

Binary fractions:
½ ¼ ⅛ 1⁄16 …

8-bit mantissa:
−1 | ½ ¼ ⅛ 1⁄16 1⁄32 1⁄64 1⁄128

4-bit exponent:
−8 4 2 1

Value = mantissa × 2^{E} . The mantissa’s binary point is immediately after the sign bit.

Binary fractions continue right of the point

Floating point uses two signed fields

The mantissa stores the signed significant fraction. The exponent stores the signed power-of-two scale.

Normalise the mantissa

For a non-zero value, a positive normalised mantissa begins 01 ; a negative one begins 10 .

If redundant 00 or 11 sign bits remain, shift the mantissa left and decrease the exponent by the same number of places.

Precision and range

More mantissa bits improve precision . More exponent bits increase range .

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Represent +6.5 in normalised floating-point form

Write 6.5 in binary

$6.5 = 110.1_2$

The whole-number part is 110 ; the 0.5 fractional part is .1 .

Move the binary point

110.1_2
$= 0.1101_2 \times 2^3$

The binary point moves three places left, so the exponent is 3 .

Write the two fields

```
Mantissa: 01101000
Exponent: 0011

Stored: 01101000 0011
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Check: 0.8125 × 8 = 6.5 .

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

Guided 1 — CHECK

Build the answer
−2 in four-bit two’s complement is 1110 .

Guided 2 — CHECK

Build the answer
Check: −0.8125 × 8 = −6.5.

Guided 3 — CHECK

Build the answer

Guided 4 — CHECK

Build the answer
$-1 \times 2^{-1} = -0.5$.

Guided 5 — CHECK

Build the answer

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

More mantissa bits improve floating-point precision.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY: 2. A small fraction: 0.15625

Use the method from I Do. Give the final value in normalised floating-point form using an eight-bit two’s-complement mantissa and a four-bit two’s-complement exponent.

Method / working:

Guided 2 — TRY: 3. Negative value: −6.5

Use the method from I Do. Give the final value in normalised floating-point form using an eight-bit two’s-complement mantissa and a four-bit two’s-complement exponent.

Method / working:

Guided 3 — TRY: 4. Write the pattern in normalised floating-point form

Use the method from I Do. Give the final value in normalised floating-point form using an eight-bit two’s-complement mantissa and a four-bit two’s-complement exponent.

Method / working:

Guided 4 — TRY: 5. Represent −0.5 in normalised floating-point form

Use the method from I Do. Give the final value in normalised floating-point form using an eight-bit two’s-complement mantissa and a four-bit two’s-complement exponent.

Method / working:

Guided 5 — TRY: Represent −15.75 in normalised floating-point form

Use the method from I Do. Give the final value in normalised floating-point form using an eight-bit two’s-complement mantissa and a four-bit two’s-complement exponent.

Method / working:

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

Represent positive values
Represent (a) 10.5 and (b) 0.375 in normalised floating-point form using an eight-bit two’s-complement mantissa and a four-bit two’s-complement exponent. Show the binary conversion and both fields. [2 each]
Adapted from the OCR 2023 conversion task: values changed; classroom marks.

Question 2

Represent negative values
Represent (a) −10.5 and (b) −0.375 in normalised floating-point form using an eight-bit two’s-complement mantissa and a four-bit two’s-complement exponent. Show the positive binary value, the mantissa negation and the final fields. [3 each]
Adapted from OCR June 2023 Q3(c): values changed; classroom marks.

Question 3

Normalise the fields
Write 00001011 0111 in normalised floating-point form using an eight-bit two’s-complement mantissa and a four-bit two’s-complement exponent. Show the shifts and exponent adjustment.
Paraphrased exam question; original pattern and marks retained. OCR June 2024, Q4(d), page 15

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 4

Diagnose the mistake
A pupil is using an eight-bit two’s-complement mantissa and a four-bit two’s-complement exponent. They change 00101000 0011 to 01010000 0011 and call it normalisation. Explain why the value changed [1], correct the exponent [1], give the complete correct pattern [1], and explain the purpose of normalisation [1].

Question 5

Handle a boundary value
Represent −0.5 in normalised floating-point form using an eight-bit two’s-complement mantissa and a four-bit two’s-complement exponent. Give the mantissa [1] and exponent [1].